

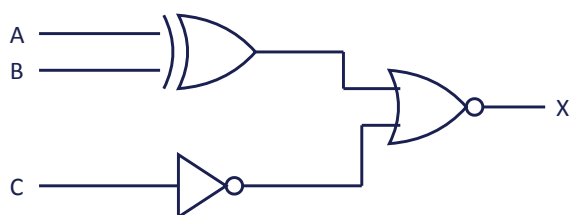
1. An aircraft warning horn **H** sounds if the landing gear **G** is *not* locked while the low-altitude sensor **L** is active, or if the test button **T** is pressed. Write the logic expression for H. [2]

H = _____

2. Draw the logic circuit for $X = (A \text{ XOR } B) \text{ NAND } (B \text{ OR } C)$. [3]

draw here — inputs A, B, C once each on the left; B must branch

3. A logic circuit is shown. Write the logic expression for output X. [2]



X = _____

4. Complete the truth table for $X = (A \text{ NAND } B) \text{ AND } (A \text{ OR } C)$. Use the working columns. [4]

A	B	C	A NAND B	A OR C	X
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

5. Write a logic expression for output X. Any logically correct expression earns the marks. [2]

X = _____

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

6. A marker is reading a hand-drawn circuit. [2]

(a) State how the marker tells a sloppily-drawn AND-family gate from an OR-family gate.

(b) Two wires cross with no dot at the crossing. State what the marker must assume.

The ladder this week — Karel, not Python. No ice-cream challenge for this lesson by design: the code you write this week is **Karel 11 Spiral and 12 StoneMason**, the two hard ones. Circuits are the paper half of the week.

Want to dig? Question 5 has no single gate that fits — you had to build it from a recipe. Now find out what that table is actually called (it outputs 1 when the inputs *agree*), and then find its opposite's day job: a single XOR gate is how a machine generates a **parity bit**, the cheapest error check in computing. Work out how one gate over eight bits can tell you that something went wrong in transmission — and, more interestingly, *what it cannot tell you*.

draft / working